ALPACA AI TRADING AGENTS HACKATHON

Alpaca AI Trading Agent

An autonomous options trading agent that runs unattended, trades real (paper) SPY iron condors on a backtested volatility signal, and logs every decision it makes.

● Live on GitHub Actions cron

Account PA3WNF5ZV8W3 · $100,000 paper

ARCHITECTURE

Analyst → Governor → Executor

Each layer has a strictly bounded role. The LLM never touches option math or picks a strike; no order reaches the broker without deterministic validation.

Analyst

Featherless LLM (DeepSeek-V3.2). Bull/bear/risk debate. **Advisory only** — logged for audit value, never gates a trade.

→

Governor

Deterministic Python. Black-Scholes IV/delta, real strike selection from the live chain, builds the exact order plan.

→

Executor

Pydantic-validated order plan → Alpaca CLI → one atomic multi-leg (MLEG) order. All legs fill together, or none do.

STRATEGY

Hybrid VRP + Skew Iron Condor

Market-neutral, defined-risk. Sell a weekly SPY iron condor only when two independent real-data signals agree.

Volatility risk premium

Enter only when implied vol (Black-Scholes, backed out of real option prices) exceeds 20-day realized vol by ≥2 points — options are priced richer than actual movement justifies.

Skew regime filter

Enter only when put-call skew isn't elevated vs. its own recent history — avoids selling premium into a market already pricing in unusual fear.

VALIDATION

52-week backtest, real Alpaca historical data

2.53
Sharpe ratio

90.9%
Win rate

1.99%
Max drawdown

SPY buy-and-hold returned +18.9% over the same window — expected, since this is a market-neutral strategy, not a directional bet. The fair comparison is risk-adjusted return, where it wins decisively.

DUE DILIGENCE

4 directional alternatives, researched and rejected

Could something beat SPY's raw return? Backtested a leveraged call diagonal, a bull risk reversal, a call-spread barbell, and momentum verticals against real data.

→ None beat SPY on a risk-adjusted basis

→ The top-ranked candidate (leveraged diagonal) was the **worst** performer once tested for real — a normal pullback erased its "safe" cushion

→ Decision: stay market-neutral, let the edge come from selectivity, not leverage

RISK MANAGEMENT

Every trade passes through hard gates first

GATE	THRESHOLD
Risk per trade	≤2% of NAV
Margin utilization	≤25% of buying power
Intraday drawdown	halt new entries at 3%
Total drawdown	kill switch (liquidate everything) at 6%
Bid-ask spread	reject any strike wider than 5% of mid
0-DTE positions	force-liquidated 3:30 PM ET

INFRASTRUCTURE

Runs unattended, on schedule, with a public audit trail

GitHub Actions cron

Every 15 minutes during US market hours. Each cycle: check circuit breakers → manage any open position → evaluate a new entry if flat and not halted.

Alpaca CLI

The only component that touches the broker. Built for agents/CI, not interactive use — satisfies the "MCP or CLI" requirement by construction.

Audit log

One structured JSON event per decision, committed back to the repo every cycle — including "nothing happened" cycles.

Live dashboard

Static site reading the repo's own audit log directly — P&L, current status, and the full decision feed.

LINKS

Thank you

- → github.com/adwik1401/alpaca-ai-trading-agent
- → WRITEUP.md — full strategy, risk gate, and architecture detail
- → dashboard/ — live audit trail and P&L